Practical 8: Model, Calculate, and Publish a RetailPulse Sales Report

Related topic: Relationships, Basic DAX, Visualizations, and Publishing Reports (Unit 3)

Name:		Roll No.:		Date:	

Note on figures and data: the tables and visuals in this report were generated from a sample RetailPulse dataset (150 sales orders, 8 products, 8 stores, 4 regions, Jan 2024 - Dec 2025) built to match the lab manual's schema. They reproduce what Power BI Desktop shows for the steps. If your workbook differs, your numbers will differ; paste your own Power BI screenshots into the marked boxes before submitting.

Objective

Create relationships between the RetailPulse tables, write basic DAX measures and a calculated column, build interactive visuals, and publish the report to the Power BI service.

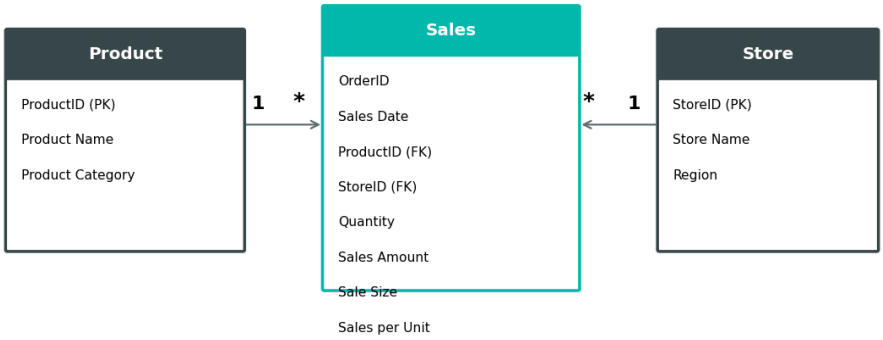
Materials Required

Power BI Desktop; RetailPulse_Analysis.pbix from Practical 7; a Power BI service account. Column names may differ slightly in your workbook, so adjust the formulas accordingly.

Procedure and Observations

Step 1 - Relationships. Open Model view. Drag Product[ProductID] onto Sales[ProductID] and Store[StoreID] onto Sales[StoreID] (or Modeling > Manage relationships > New). Set **Cardinality: One to many (1:*)**, **Cross-filter direction: Single**, Make this relationship active: ticked.

Model view: one-to-many, single direction (dimension filters fact)



Step 2 - Measures. Home > New measure, and enter:

```
Total Sales = SUM(Sales[Sales Amount])
Total Quantity = SUM(Sales[Quantity])
Average Order Value = DIVIDE([Total Sales], DISTINCTCOUNT(Sales[OrderID]))
```

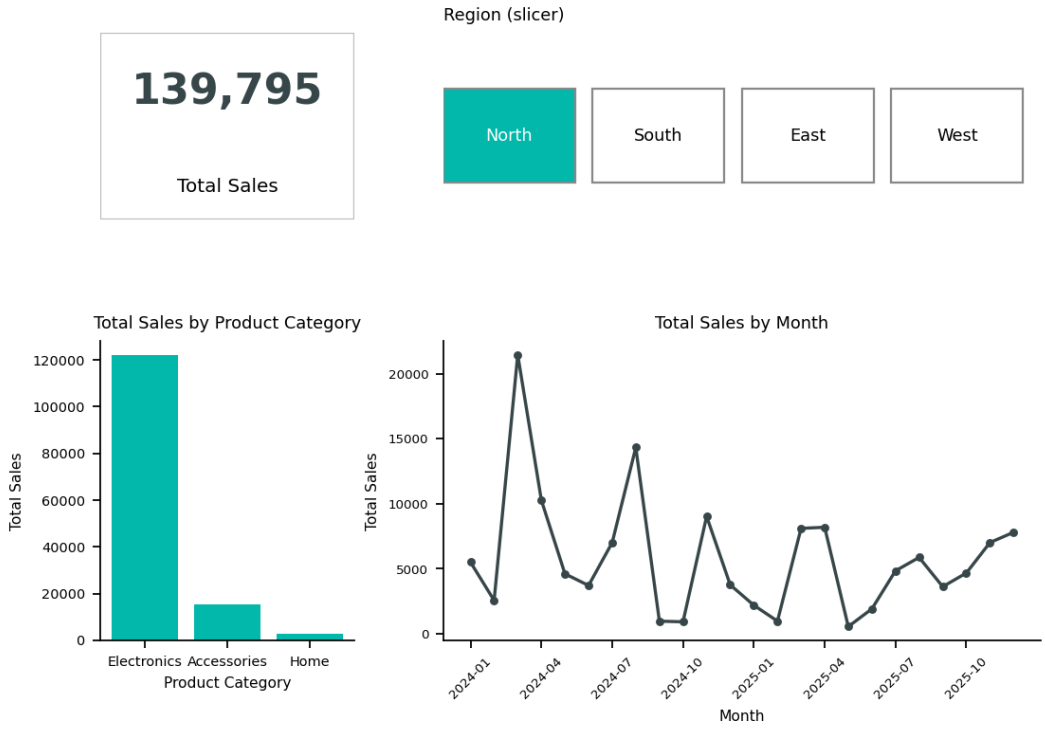
Step 3 - Calculated column. Select the Sales table > New column:

```
Sales per Unit = DIVIDE(Sales[Sales Amount], Sales[Quantity])
```

Sample output of the calculated column (a row with Quantity = 0 shows blank instead of an error because DIVIDE handles division by zero):

OrderID	Sales Amount	Quantity	Sales per Unit
ORD1132	2,675.33	3	891.78
ORD1080	276.55	5	55.31
ORD1076	121.32	3	40.44
ORD1141	2,027.44	5	405.49
ORD1055	80.88	2	40.44
ORD1042	319.62	4	79.91

Step 4 - Report page. Insert: *Card* (Total Sales); *Clustered bar/column chart* (Axis = Product[Product Category], Values = Total Sales); *Line chart* (Axis = Sales[Sales Date] by Year-Quarter-Month or month, Values = Total Sales); *Slicer* (Field = Store[Region]).



Step 5 - Interaction test. Click the Electronics bar: the card, line chart and slicer-linked visuals filter to Electronics. Then choose a Region in the slicer: every visual updates. Add titles and axis labels through Format > General > Title and Format > X/Y axis > Title.

Effect of the Region slicer on the measures (sample data):

Region	Total Sales	Total Quantity	Orders	Average Order Value
East	47,708.85	125	39	1,223.30
North	27,753.84	83	34	816.29
South	30,294.89	105	36	841.52
West	34,037.14	122	41	830.17
All (no slicer)	139,794.72	435	150	931.96

Step 6 - Publish. Home > Publish > sign in > choose a workspace (e.g. My workspace) > Select. Open the link "Open RetailPulse_Analysis.pbix in Power BI". Share via the Share button, entering the instructor's e-mail; others can open it only if they have workspace access or the report is shared with them (and a Pro/PPU licence where required).

Paste your Power BI screenshot here: Model view with relationships

Paste your Power BI screenshot here: Published report in the Power BI service

Result / Expected Outcome Check

• Model view shows Product and Store each linked to Sales with the 1 and * markers on the correct ends.

• Measures return values: Total Sales = 139,794.72; Total Quantity = 435; Average Order Value = 931.96 (sample data).

• Measures change when the Region slicer changes; the Total Sales card equals the sum of the bars in the category chart.

• The report is visible in the chosen workspace; others can open it only if given access.

Viva-Voce Questions and Answers

a) Why is a one-to-many relationship from a dimension table to a fact table preferred?

A dimension table (Product, Store) holds each key once with descriptive attributes, while the fact table (Sales) repeats the key for every transaction. A 1:* relationship lets a filter on the dimension flow to the fact rows unambiguously, avoids duplicated or ambiguous results, saves memory, and is the basis of the star schema that Power BI's engine is optimised for.

b) What is the difference between a measure and a calculated column?

A **calculated column** is computed row by row when the data loads/refreshes and stored in the model (uses memory); it can be used in slicers, axes and filters. A **measure** is computed at query time in the current filter context and is not stored; it reacts to slicers and cross-filtering and is used for aggregations such as totals and ratios.

c) Why is the DIVIDE function used instead of the / operator?

DIVIDE handles division by zero or blank safely: it returns BLANK (or a chosen alternate result) instead of an error or infinity, so visuals do not break. It is also concise for the alternate-result argument.

d) Which visual would you choose to show a trend over time, and why?

A **line chart**, because the connected points on a continuous time axis make direction, seasonality and rate of change easy to see; bars are better for comparing categories.

e) What must be in place before another person can view your published report?

The report must be published to a workspace, and the person must have access to it (workspace role or direct share/app permission) and an appropriate licence (Pro or Premium Per User, unless the workspace is on Premium capacity). If the report uses row-level security, their role assignment must also be set.